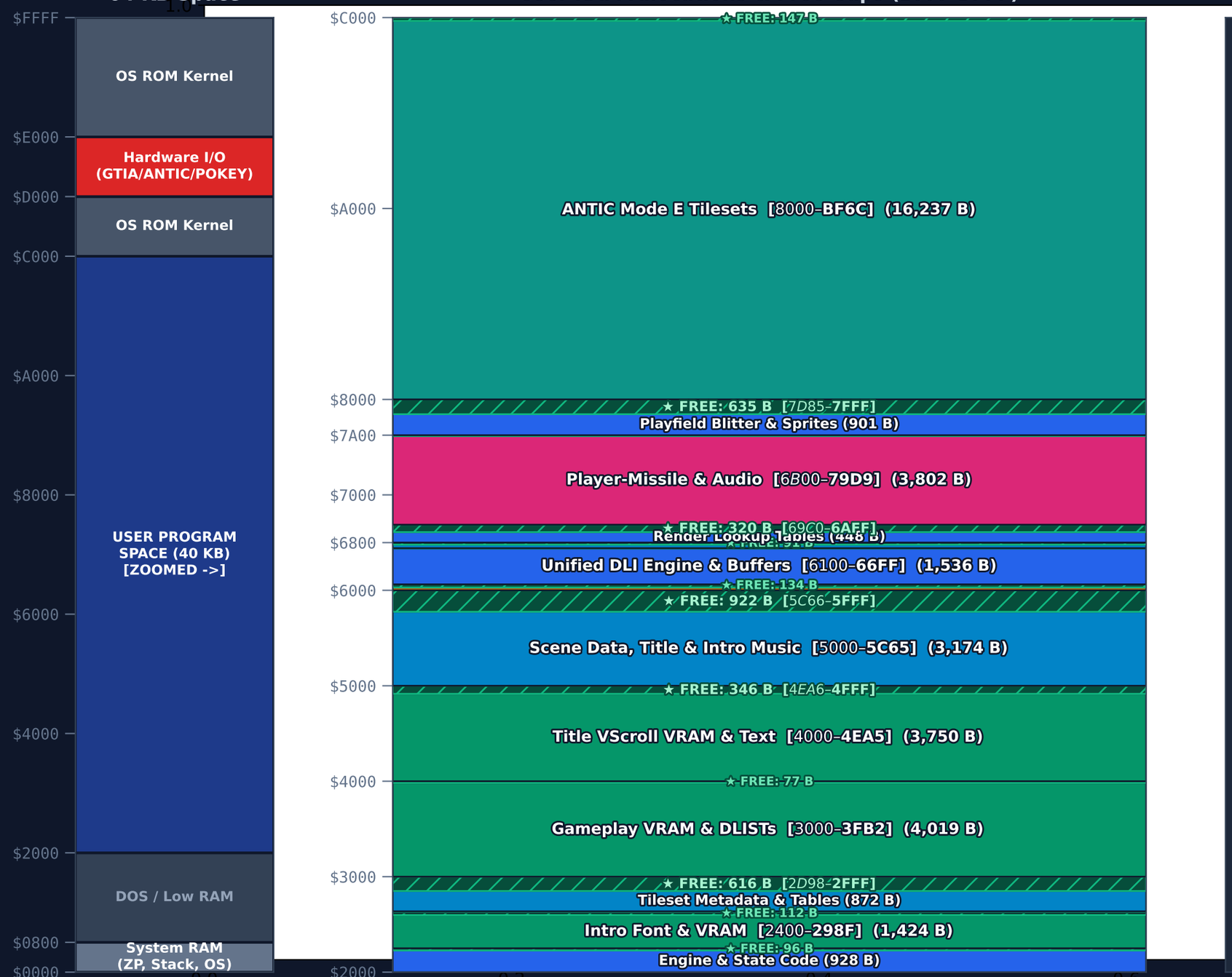
ATARI 8-BIT MEMORY MAP — CALL OF THE WINTERMON

File: call_of_the_wintermon.xex | Target: Atari 800XL / 65XE (MOS 6502) | RAM: 64 KB | User RAM: 2000–BFFF (40 KB)



MEMORY UTILIZATION SUMMARY

User RAM Space (2000–BFFF):	**40,960 Bytes (40.0 KB)**
Allocated Memory:	**37,392 Bytes (91.3%)**
Total Free Headroom:	**3,568 Bytes (8.7%)**
Largest Free Block:	**922 Bytes (5C66–5FFF)**
Free Zero Page RAM:	**63 Bytes (C1–FF)**

CATEGORY LEGEND

CODE	Game Engine ($2000), DLI Engine ($6100), Render ($7A00)
GRAPHICS	Playfield VRAM (3000)&*Title/Scroller*(4000)
BITMAPS	ANTIC Mode E Tileset Sheets ($8000)
DLIST	Display Lists (Game $3F00, Title $6000, Intro $2A00)
PMG / AUDIO	Player-Missiles (6800)&*AudioPlayer*(7000)
METADATA	Scene SoA Tables, Title Code & Intro Music ($5000)
FREE RAM	Available unallocated memory gaps
HARDWARE / ROM	GTIA/ANTIC/POKEY (*D*000)&*OSROM*(C000+)

TOP FREE MEMORY REGIONS (HEADROOM)

1. 5C66–**5FFF (922 Bytes / 0.90 KB)**
   Free expansion headroom in 5000*blockbeforeTitleDLIST*(6000)
2. 7D85–**7FFF (635 Bytes / 0.62 KB)**
   Free headroom in 7000*blockbeforeANTICModeETilesets*(8000)
3. 2D98–**2FFF (616 Bytes / 0.60 KB)**
   Low RAM gap before Gameplay VRAM ($3000) — vacated by scenes data
4. 4EA6–**4FFF (346 Bytes / 0.34 KB)**
   Free space after Title Scroller before $5000 boundary
5. 69C0–**6AFF (320 Bytes / 0.31 KB)**
   Free headroom in PMG block ($6800-$6AFF) before PMG missiles ($6B00)

ZERO PAGE BREAKDOWN (00–FF)

00–**7F (128 B)**	Reserved for Atari OS Kernel
80–**8F (16 B)**	Engine temporaries & state machine
90–**99 (10 B)**	POKEY Audio Player base pointers
9A–**B7 (30 B)**	Scene background renderer pointers
B8–**C0 (9 B)**	DLI engine registers & pattern pointers
C1–**FF (63 B)**	★ 100% FREE for Entities, AI, Physics